

QNS-CD-1.0 ? Quantum Node Signal coding design

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Status: LIVE fabric coding design (2026-09-10). Not a Softwares-tab product. Not a FragGate slug. Not a fleet-completeness claim.

Aziel Eliab • coding design • photon QNS1 1.3 • local qnsd only

Quantum Node Signal
Packet-transfer coding design for photon vias

QNS-CD-1.0 10 September 2026 Author: Aziel Eliab
Companion: QNM-BUILD-1.0 / AIH-WP-1.3 (qnm-node)
Hub / Interface law (unchanged): AIH-WP-1.1
Implementation: local process qnsd in <https://github.com/AzielEliab/qnm-node>
Public Worker: cites only. Never a remote wipe or control plane.

0. Sentence

A Quantum Node Signal is a folded, airlocked, fact-hashed frame that may travel a photon via on local qnsd. The public Worker names the law. It does not carry the via.

1. Claim

QNM-BUILD-1.0 already said: local process ON, public rollup OFF, GET /v1/mesh never enables. AIH-WP-1.3 already said: pair_id is medium-independent; bearer is hop-only; forward only along existing spiderweb edges after APG.

This paper is the ****coding design**** for the packet that rides those edges: photon ****QNS1 1.3****. The daemon is ****qnsd****, a local module of qnm-node. It binds ****127.0.0.1**** only.

aziel-runtime already admits Worker traffic through AZPIPE (AP-WP-0.2), SweepGate (SG-WP-0.1), APG on the local node, and ChainLock (CL-WP-0.4). qnsd uses ****the same laws**** on the loopback hop. Changing product names does not change the list.

2. What it is not

- ? Not a Softwares-tab product. Do not add slug qns, qnsd, or photon.
- ? Not a FragGate engine. No fraggate_call { slug: "qns" }.
- ? Not qubit hardware, Bell-pair physics, or a quantum computer claim.
- ? Not a public proxy of 127.0.0.1. The Worker must not emit, via, wipe, scorch, arm, or forward on behalf of a remote caller.
- ? Not a remote wipe or control plane. EmbryoLock stays stub.
- ? Not Node Gate / IP panel / login mesh / publish path.
- ? Not AZMail's product-local ring. Not AnonBroadcast as a catalog product.
- ? Not a substitute for AZPIPE, SweepGate, APG, or ChainLock ? it cites them.

3. Two planes (unchanged)

Plane	Default	Law
A ? local qnsd / qnm-node	Process ON. Via attempt.	Bind 127.0.0.1. APG on every ingress. Photon via is hop-only.
B ? public Worker	Cite only. GET /v1/qns.	No emit. No proxy. GET

4. Photon QNS1 1.3

Magic QNS1. Version 1.3. Canonical JSON (sorted keys, tight separators). Body hash h is SHA-256 of the frame with h omitted. Fact hash fh is SHA-256 of the clipped fact (?160). QNM and AZPIPE already cite fh. Remote body does not enter STM.

```
{
  magic: "QNS1",
  v: "1.3",
  via: "photon",
  pair_id: "<AIH-WP-1.3 pair_id>",
  hop: { n: 0, max: 8, seen: [] },
  fh: "<sha256 fact>",
  h: "<sha256 frame minus h>",
  isolate: false,
  pull: false,
  bridges: 2,
  inner: null
}
```

via is hop-only (AIH-WP-1.3). Any currently enabled declared bearer may carry the next hop. The pair does not store a required medium. Wi-Fi dying is path gone, bind remains.

hop.max default 8. seen is a hash list. Loops drop. Over-length paths drop. pull stays false. bridges=2. isolate=true on poison / tamper.

inner is the AZPIPE-admitted folded body, or null on a refuse envelope. Refuse envelopes stop at the hop that closed. They do not carry inner into memory.

5. Same laws locally

Inbound hop list (AP-WP-0.2), on qnsd:

frag ? sweep ? fold ? static ? fold ? entry ? frag ? via

frag. DecisionGATE flags. Empty or ungrounded inbound refuses before SweepGate spends work.

sweep. SweepGate airlock (SG-WP-0.1). Poison, malware-class marks, block-keys, off-origin when inbound and untrusted. Isolate and refuse. Do not merge.

fold / static. fld3-wire until foldlock.py sits beside the hop. Freeze hashes the folded bytes. Second fold after canonicalization.

entry. ChainLock stamp (CL-WP-0.4). Fact-bearing card. Hash-only cards refuse.

via. Emit only along an existing pair edge, APG pass, hop bearer enabled, hop_max respected, node not isolated / PHOENIX-LOCK / scorched.

APG (qnm-node): clean pass; reanswer-without-cite refuse; flood refuse; unknown slug or stub-as-live refuse; credential/login/cookie refuse; poison/tamper isolate + spend ID + drop tethers. Poison is refused, not interpreted.

Outbound is the reverse so a dirty toolkit result is swept before it rides a via.

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6. Local API (qnsd ? 127.0.0.1 only)

Method	Path	Act
GET	/local/qns/status	Photon via posture. Not completeness. Not QNM-S.
GET	/local/qns/cite	QNS-CD-1.0 + QNS1 1.3.
GET	/local/qns/outbox	Visible queue. Operator keeps the file.
POST	/local/qns/admit	APG then admit inbound QNS1 frame.
POST	/local/qns/via	Emit along an existing pair edge.
POST	/local/qns/cut	Drop one outbox item.

These paths live on local qnsd in AzielEliab/qnm-node. They are ****cited**** by GET /v1/qns on aziel-runtime. The public Worker never fetches them, never proxies them, and never accepts emit / via / wipe / control bodies.

7. Public Worker (this package)

GET /v1/qns ? cite JSON (author, spec, photon, local API table, laws).
HEAD /v1/qns ? headers only.
POST /v1/qns ? refuse QNS-CITE-ONLY.
POST /v1/qns/via | /emit | /wipe | /arm ? refuse QNS-NO-PROXY.

Every GET /v1/software card and the mesh kernel extra carry a stable pointer:

```
qns_cd: {  
  spec: "QNS-CD-1.0",  
  local: "https://github.com/AzielEliab/qnm-node",  
  note: "Photon vias on local qnsd; Worker cites only"  
}
```

Do not add QNS as a Softwares-tab product. extras[] may show the pointer on the mesh kernel card. FragGate remains the door. Mesh remains the rollup.

Mesh status / skill notes cite QNS-CD-1.0 as the packet-transfer coding design (companion to QNM-BUILD-1.0 / AIH-WP-1.3). Hub / Interface stays AIH-WP-1.1.

8. Cross-map

The pointer is the same object on every live product, EmbryoLock stub, and the mesh kernel extra. Catalog GET stays full (RL-WP-0.1-runtime). Soft caps apply only to expensive fan-out. Packed catalog cost path is unchanged.

9. Never

? Do not enable mesh with a GET.
? Do not bind qnsd off loopback in default config.
? Do not proxy local via emit from the public Worker.
? Do not add a Node Gate to ?see the photon?
? Do not invent qubit claims or Bell-pair physics.
? Do not host EmbryoLock wipe on this door.
? Do not list QNS on the Softwares tab.
? Do not treat a site ping of /v1/qns as a via.

10. Close tests

? SUITE_DESIGNS includes QNS-CD-1.0; skill / llms / cite / sitemap name it.
? Every /v1/software card has qns_cd.spec === "QNS-CD-1.0".
? Mesh extra and GET /v1/mesh cite QNS-CD-1.0.
? GET /v1/qns is 200, cite-only, public_proxy false, emit false.
? POST /v1/qns and /v1/qns/via refuse. No 127.0.0.1 fetch in the Worker.
? No software[] slug qns / qnsd / photon.
? GET /v1/mesh remains enabled:false on a fresh isolate.

identity: Aziel Eliab only.

If the files hold, the name was never the point.

Packet-transfer coding design • local qnsd • Worker cites only • 2026-09-10
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